AGA KHAN UNIVERSITY EXAMINATION BOARD

Notes from E-Marking Centre on SSC-I Chemistry Annual Examination 2023

Official Examiner Observations, Marking Schemes, and Performance Analysis

Introduction: This document has been produced for the teachers and candidates of Secondary School Certificate (SSC) Part I Chemistry. It contains comments on candidates' responses to the 2023 SSC-I Examination indicating the quality of the responses and highlighting their relative strengths and weaknesses.

Question No. 1 (Atomic Structure & Ion Drawing)

3 Marks | Cognitive Level: Application | SLO 2.1.3

Prompt: Draw the structure of an ion with Charge of -3, Atomic number = 15, Nucleon number = 31.

Examiner Feedback: Overall, candidates demonstrated a good understanding of ion formation, particularly for the P3- ion. They accurately depicted the 3 electron shells (2, 8, 8) and nuclear composition (15 protons, 16 neutrons).

Better Responses: Better responses showed clarity in illustrating electron, proton, and neutron counts with a -3 charge and correct electronic arrangement before and after ion formation.

Weaker Responses: Weaker responses displayed misconceptions, incorrectly assuming equal numbers of protons, neutrons, and electrons in P3- or omitting the negative bracket charge.

Question No. 2 (Physical Properties of Halogens)

3 Marks | Cognitive Level: Knowledge | SLO 3.1.6

Prompt: Complete blanks for Cl (Odour), Br (Physical state at RT), and I (Colour).

Examiner Feedback: Overall performance in understanding general physical properties of halogens was satisfactory.

Better Responses: Better responses accurately stated pungent/choking odour for chlorine, reddish-brown liquid state for bromine, and purple/dark grey-black colour for iodine.

Weaker Responses: Weaker responses mistook chlorine for an odourless gas, or gave incorrect colours and states for bromine and iodine.

Question No. 3 (Diffusion of Gases)

3 Marks | Cognitive Level: Understanding | SLO 5.2.1

Prompt: Heating NH_4Cl produces NH_3 and HCl gases moving along the tube. (a) Name the process. (b) Which gas travels faster and why?

Examiner Feedback: Candidates performed moderately well in identifying the process and gas, but struggled to give a complete mathematical reason based on molar mass.

Better Responses: Better responses identified diffusion and explained that ammonia has a lower molar mass (17 g/mol) than HCl (36.5 g/mol), thus diffusing faster by Graham's law.

Weaker Responses: Weaker responses confused diffusion with sublimation or decomposition, or claimed HCl diffuses faster.

Question No. 4 (Combustion of Magnesium)

4 Marks | Cognitive Level: Understanding | SLO 8.2.4

Prompt: Mg ribbon burns with dazzling light to produce white MgO. (a) Identify Mg. (b) Write balanced chemical equation. (c) Identify bond type.

Examiner Feedback: Most candidates correctly identified magnesium and stated ionic bonding. Writing the balanced chemical equation was the main discriminator.

Better Responses: Better responses wrote: 2Mg(s) + O2(g) -> 2MgO(s) and correctly identified ionic bonding.

Weaker Responses: Weaker responses wrote unbalanced equations such as Mg + O -> MgO or Mg + O2 -> MgO2, failing to represent diatomic oxygen.

Question No. 5a (Empirical & Molecular Formula)

6 Marks | Cognitive Level: Application | SLO 1.4.3

Prompt: Compound Y: 80% C, 20% H, Molar mass = 30.07 g/mol. Calculate empirical and molecular formula.

Examiner Feedback: Candidates performed well in presenting the necessary steps for calculating moles and atomic ratios.

Better Responses: Better responses showed: C = 80/12 = 6.67, H = 20/1 = 20. Ratio C:H = 1:3 -> Empirical formula CH3. Empirical mass = 15. n = 30.07/15 = 2 -> Molecular formula C2H6.

Weaker Responses: Weaker responses used percentages directly without converting to moles, or divided by atomic numbers rather than atomic masses.

Question No. 5b (Crystallisation of Copper Sulphate)

6 Marks | Cognitive Level: Application | SLO 6.6.2

Prompt: Obtaining solid CuSO4 from aqueous solution. (i) Why filtration fails. (ii) Alternate method. (iii) Three-step procedure.

Examiner Feedback: Candidates understood why filtration fails (homogeneous solution), but weaker responses gave inaccurate practical steps.

Better Responses: Better responses explained that dissolved copper sulphate passes through filter pores, named crystallisation, and detailed heating to saturation, cooling for crystal growth, and filtering/drying crystals.

Weaker Responses: Weaker responses suggested boiling to dryness or did not mention cooling for crystal formation.

Question No. 6a (Ionic Bonding in CaCl2)

6 Marks | Cognitive Level: Understanding | SLO 4.2.4

Prompt: Dot and cross structure of CaCl2: (i) Identify bond. (ii) Role of each atom. (iii) Three characteristics.

Examiner Feedback: Most candidates correctly identified ionic bonding in CaCl2.

Better Responses: Better responses explained that calcium loses 2 electrons to form Ca2+ and two chlorine atoms each gain 1 electron to form Cl-, and cited high melting point, electrical conductivity in molten/aqueous state, and water solubility.

Weaker Responses: Weaker responses incorrectly claimed electron sharing or described single covalent bonds.

Question No. 6b (Electrolysis of Brine)

6 Marks | Cognitive Level: Understanding | SLO 7.5.2

Prompt: Define electrolysis and describe electrolysis of concentrated NaCl solution (brine) with equations.

Examiner Feedback: Most candidates defined electrolysis well and knew chlorine and hydrogen are produced.

Better Responses: Better responses defined electrolysis as chemical decomposition by electric current, gave anode oxidation: 2Cl- -> Cl2 + 2e-, cathode reduction: 2H+ + 2e- -> H2, and explained NaOH formation in solution.

Weaker Responses: Weaker responses confused brine electrolysis with Down's cell (molten NaCl), claiming sodium metal is produced at the cathode.